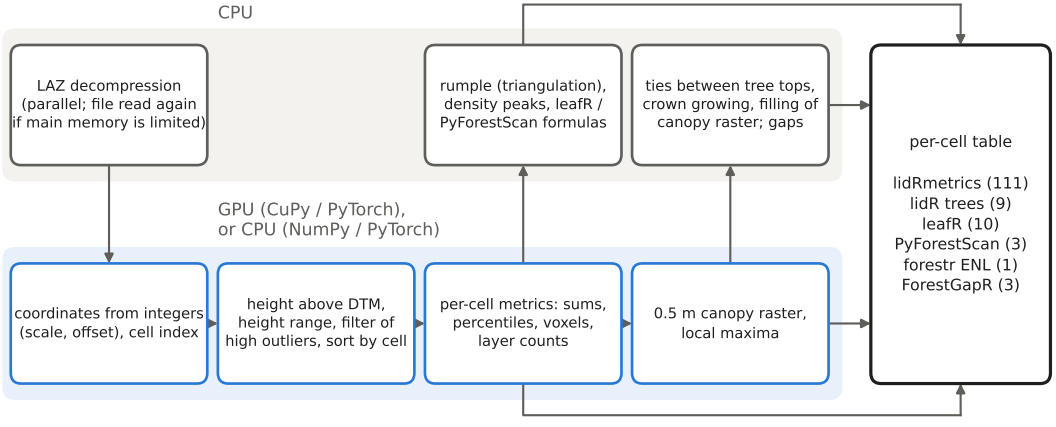with a memory limit, the steps of the lower row run on blocks of whole rows of cells